completely limited by the sampling aperture, and not the sample spacing. Put another way, the sampling aperture has acted as an antialias filter, allowing lossless sampling to take place.

In comparison, the MTF curve in (d) shows that *both* processes are limiting the resolution of this system. The high-frequency fall-off of the MTF curve represents information lost due to the *sampling aperture*. Since the MTF curve has not dropped to zero before a frequency of 0.5, there is also information lost during sampling, a result of the finite *sample spacing*. Which is limiting the resolution more? It is difficult to answer this question with a number, since they degrade the image in different ways. Suffice it to say that the resolution in the perfect detector (example 1) is mostly limited by the sample spacing.

While these concepts may seem difficult, they reduce to a very simple rule for practical usage. Consider a system with some 10%-90% edge response distance, for example 1 mm. If the sample spacing is greater than 1 mm (there is less than one sample along the edge), the system will be limited by the *sample spacing*. If the sample spacing is less than 0.33 mm (there are more than 3 samples along the edge), the resolution will be limited by the *sampling aperture*. When a system has 1-3 samples per edge, it will be limited by both factors.

Signal-to-Noise Ratio

An object is visible in an image because it has a different brightness than its surroundings. That is, the contrast of the object (i.e., the signal) must overcome the image noise. This can be broken into two classes: limitations of the *eye*, and limitations of the *data*.

Figure 25-7 illustrates an experiment to measure the eye's ability to detect weak signals. Depending on the observation conditions, the human eye can detect a minimum contrast of 0.5% to 5%. In other words, humans can distinguish about 20 to 200 shades of gray between the blackest black and the whitest white. The exact number depends on a variety of factors, such

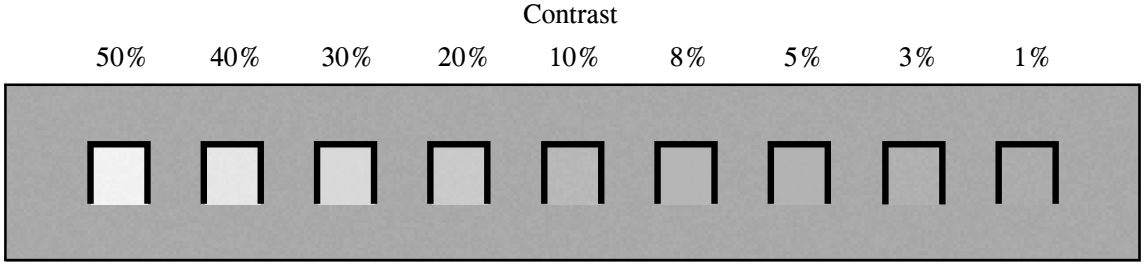

FIGURE 25-7
Contrast detection. The human eye can detect a minimum contrast of about 0.5 to 5%, depending on the observation conditions. 100% contrast is the difference between pure black and pure white.